Fall DRP Final - Vignesh Nydhruva

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The purpose of this document is to provide a small glimpse into algebraic number theory. We begin with a simple example. Consider $\mathbb{Z}[\sqrt{d}]$, the ring of all $\mathbb{Z}$-linear combinations of 1 and $\sqrt{d}$, where $d > 1$ is squarefree in $\mathbb{Z}$. We want to show that $\mathbb{Z}[\sqrt{d}]$ has infinitely many units. $\mathbb{Z}[\sqrt{d}]$ is equipped with the norm function N, defined so that $N(a + b\sqrt{d}) = (a + b\sqrt{d})(a - b\sqrt{d})$. So, the invertible elements of $\mathbb{Z}[\sqrt{d}]$ correspond to the elements of $\mathbb{Z}[\sqrt{d}]$ of norm ± 1. It suffices to show there are infinitely many elements of norm 1. Thus we obtain the equation $a^2 - db^2 = 1$. This reduces to finding the number of solutions to Pell's equation, with the given conditions. By [3], a solution $a_1 + b_1\sqrt{d}$ exists, and we may exponentiate $(a_1 + b_1\sqrt{d})^n$ to find other solutions for n a positive integer. So we have found infinitely many solutions to our Pell equation, and so there are infinitely many elements in $\mathbb{Z}[\sqrt{d}]$ that are units.

An *algebraic number field* K is a finite extension of $\mathbb{Q}$. Recall that an element $a \in K$ is called *algebraic* if it is a root of a monic polynomial with coefficients in $\mathbb{Q}$; we say that $a \in K$ is an *algebraic integer* if it is algebraic and the mentioned polynomial has integral coefficients. Now we introduce some notation and definitions given in [2] and [4]. We write $\mathcal{O}_K$ to denote the ring of algebraic integers (or more simply, the *ring of integers*) of the number field K. We say that the ring of integers is the integral closure of $\mathbb{Z}$ in K. For instance, the trivial extension $\mathbb{Q}/\mathbb{Q}$ has $\mathcal{O}_K$ to be $\mathbb{Z}$ and the number field $K = \mathbb{Q}(\sqrt{3})$ has $\mathcal{O}_K = \mathbb{Z}(\sqrt{3})$. Now, for an integral domain R, let K be its field of fractions. Then define a *fractional ideal* I of R to be an R-submodule of K so that for some nonzero $r \in R$, we have $rI \subset R$. For example, let $R = \mathbb{Z}$, so its field of fractions is $K = \mathbb{Q}$, and consider $I = \frac{3}{7}\mathbb{Z} \subset K$. We can pick $r = 7 \neq 0$ in $\mathbb{Z}$ and we have $rI = 3\mathbb{Z} \subset R$, and so I is a fractional ideal of $\mathbb{Z}$. Now for a number field K, define J_K to be the group of fractional ideals of $\mathcal{O}_K$ and let P_K be the subgroup of J_K consisting solely of its principal ideals. We define the *class group* of the number field K to be the quotient group J_K/P_K and say that the *class number* is the order of the class group. We now make a few concluding definitions. Suppose K is a number field and I is a nonzero ideal of $\mathcal{O}_K$. Then the *norm* of I is defined to be $[\mathcal{O}_K : I] = |\mathcal{O}_K/I|$. Now for a number field K, the *Minkowski bound* states that each class in J_K/P_K contains an integral ideal with norm (as previously defined) at most

$$\sqrt{|\operatorname{disc}(K)|}\left(\frac{4}{\pi}\right)^{r_2}\frac{n!}{n^n}$$

where $n = [K : \mathbb{Q}]$, $\operatorname{disc}(K)$ is the discriminant of the number field K (i.e. if $K = \mathbb{Q}(\alpha)$, then $\operatorname{disc}(K)$ is the discriminant $b^2 - 4ac$ of the minimal polynomial $m(x) \in \mathbb{Q}[x]$ of α), and $r_2 = \frac{n - r_1}{2}$, where r_1 is the number of real embeddings.

We now calculate a class group using a method outlined in [2]. Suppose $K = \mathbb{Q}(\sqrt{7})$. Then $r_1 = 2$, $n = 2$, so $r_2 = 0$. Since the minimal polynomial of $m(x) := \sqrt{7}$ is $x^2 - 7$, thus the discriminant is $4 \cdot 7$, so $|\operatorname{disc}(K)| = 28$. Thus the Minkowski bound is $\sqrt{28} \cdot (4/\pi)^0 \cdot (2!/2^2) \approx 2.65$. Reducing modulo 2 gives that, by [1], $m(x)$ ramifies as it is the square of the linear polynomial $x+1$ over $\mathbb{F}_2$. By the Minkowski bound, each ideal class in the class group is given by an ideal of norm at most 2. It can be shown that factoring the ideal (2) as a product of ideals in $\mathbb{Q}(\sqrt{7})$ shows that the class group is $\mathbb{Z}/2\mathbb{Z}$.

References

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